Preliminary Sample-Size Analysis for CLAPS

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Abstract

CLAPS asks whether affectedness, how strongly an event affects the person or thing it happens to, matters more for the acceptability of passive sentences than of active ones, and whether that pattern holds across languages. This report evaluates by simulation what the planned sample of 80 participants per language can be expected to deliver. Studies were simulated from the planned ordinal model fitted to the pilot data in English, Turkish and Norwegian, with the effects drawn from each language's pilot posterior, and each was analysed as the real data will be. The design has about 72 verbs per language, with agent gender counterbalanced across participants. At 80 participants, the primary hypothesis (H1), that the affectedness slope is steeper for passives than for actives, reaches a directional Bayes factor of 10 in 90% of simulated English studies and 97% of Norwegian ones. Turkish, whose pilot effect is weakest, reaches it in 57%. The secondary hypothesis (H2), that the slope is also steeper for passives than for pseudo-passives, reaches a Bayes factor of 10 in 95% of English studies, while the Turkish pilot shows the reverse pattern and the simulations inherit it. Affectedness is a property of the verb, so the verbs bound the precision of every test, and beyond 80 participants the rates barely change. The pooled cross-linguistic analysis, planned as a synthesis with no confirmatory claim, reaches a Bayes factor of 10 on H1 in 57% of simulated studies with the three pilot languages, at 50 to 100 participants per language. With the 20 languages the collaborators expect, the seven measured so far among them, it does so in 89% of studies on H1 and, across the 15 with a pseudo-passive, in 90% on H2, if further languages differ as much as the measured ones.

Background

CLAPS (A Crosslinguistic Approach to Passive Semantics) asks whether the acceptability of passive and related sentence types tracks the semantic affectedness of the patient, whether that relationship differs by sentence type and whether it holds across languages. Affectedness, the degree to which a patient or theme undergoes a change of state or is causally impinged upon by the action, is a core dimension of argument structure (Beavers, 2011; Dowty, 1991). It has long been argued to be the semantic heart of the passive, which carries the meaning that the surface subject is in a state or circumstance characterised by the agent having acted upon it (Pinker et al., 1987). Acceptability-judgement and comprehension studies with adults in several languages support the view that passive acceptability is semantically structured (Ambridge et al., 2016; Ambridge et al., 2023; Aryawibawa & Ambridge, 2018; Bidgood et al., 2020; Darmasetiyawan & Ambridge, 2022; Liu & Ambridge, 2021).

The evidence to date spans English, Indonesian, Balinese, Hebrew and Mandarin. A Bayesian meta-analytic synthesis of those studies finds that affectedness raises the acceptability of passives but not of pseudo-passives (Ambridge et al., 2023). In most of the five languages, it raises passive acceptability more than active acceptability, although the size of that advantage varies considerably and in Balinese it runs the other way. CLAPS adds Turkish and Norwegian, for which its pilot supplies estimates.

Each language's data will be collected by its own team, at a sample size fixed in advance. This report, prepared for the study's Stage 1 Registered Report, evaluates what the planned sample can be expected to deliver. It does so through a design analysis (Schönbrodt & Wagenmakers, 2018) whose simulated effects come from the CLAPS pilot.

The Model and the Predictions

Each acceptability judgement is a rating from 1 to 7. The ratings are modelled with a Bayesian cumulative-logit mixed-effects model, an ordinal model that respects the discreteness, ordering and bounded range of rating data (Bürkner & Vuorre, 2019). Treating the same ratings as continuous biases the regression coefficients and inflates error rates where floor or ceiling effects are present (Liddell & Kruschke, 2018), and modelling them cumulatively recovers psycholinguistic effects more accurately (Taylor et al., 2023; Veríssimo, 2021).

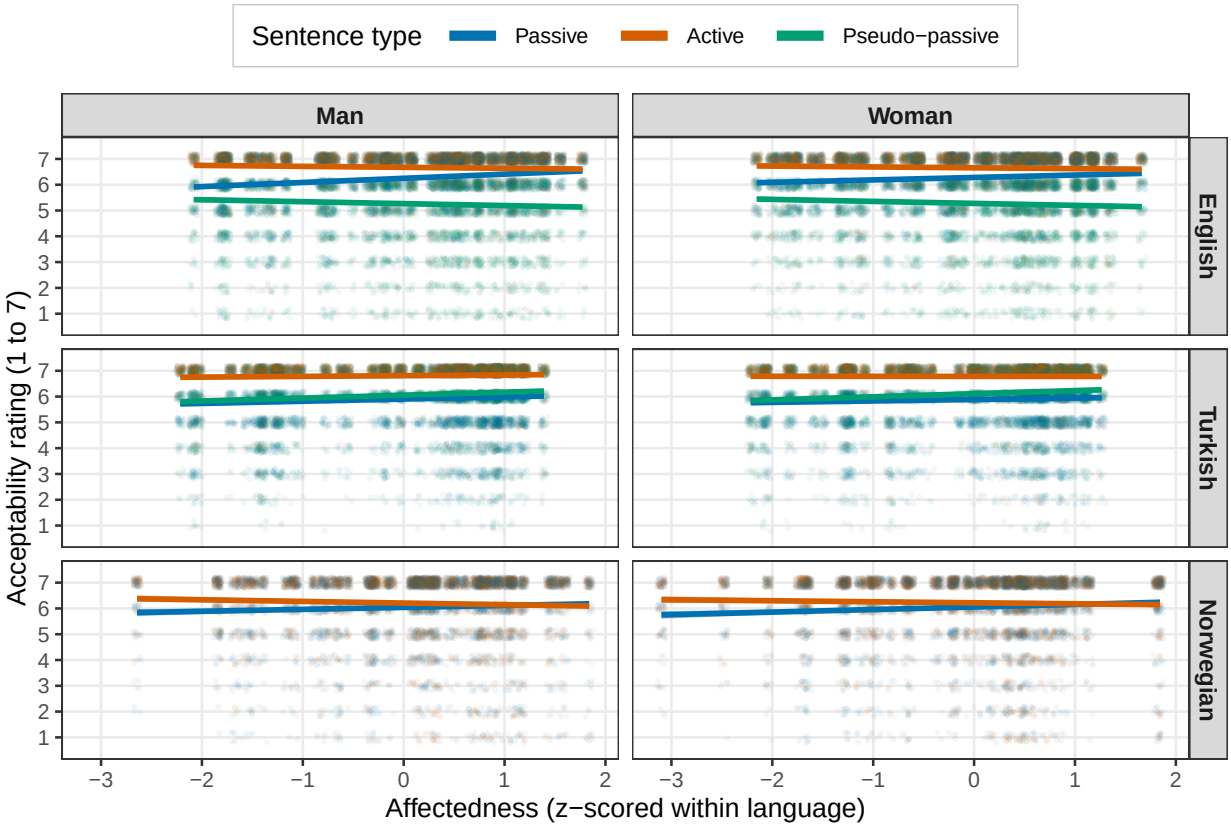
The model predicts acceptability from sentence type, affectedness and their interaction. Each stimulus appears as an active, a passive and a pseudo-passive. In English, for example, these are "The woman pushed the man", "The man was pushed by the woman" and "It was the man that the woman pushed", the last a cleft that puts the patient first without the passive's morphology. Each

sentence type is treated as the same construction in every language that has it. The passive is the reference level, and the active and the pseudo-passive are compared with it.

The pilot, whose sample Appendix A describes, shows the expected pattern in English and Norwegian (Figure 1). In both, the affectedness slope is positive and steepest for the passive, and flat or slightly negative for the active, as it is for the English pseudo-passive. In Turkish, the passive slope is also steeper than the active slope, though by less, while the pseudo-passive slope is at least as steep as the passive slope. The trends look much the same for male and female agents, although in English the active-by-affectedness interaction differs between agent genders. The design therefore balances the two agent genders across participants (Appendix B).

Figure 1

Raw Pilot Acceptability by Affectedness, Sentence Type, Agent Gender and Language



Note. Panels are by language across rows and by agent gender across columns. Points are individual pilot ratings, jittered, and lines are per-sentence-type linear trends with 95% confidence bands, which ignore the clustering of ratings within participants and verbs and so understate the uncertainty. Affectedness is the score normed for each agent-gender version, z-scored within language, and the model uses one score per verb, the mean of the two. The Norwegian materials include no pseudo-passive.

The study tests two hypotheses. H1, the primary hypothesis, holds that the affectedness slope is steeper for passives than for actives, so that the active-by-affectedness interaction is negative. H2, the secondary hypothesis, holds that the slope is also steeper for passives than for pseudo-passives, so that the pseudo-passive-by-affectedness interaction is negative. The meta-analysis (Ambridge et al., 2023) and the English pilot both put that interaction below zero. The passive slope itself carries no prediction. Affectedness might raise or lower the acceptability of sentences in general, and the theory requires only that it matter more for passives, so each test compares the passive slope with another slope, whatever their signs. H1 is tested in every language, and H2 in every language with a pseudo-passive, which the Norwegian materials do not include.

Design-Analysis Method

The design analysis simulates many studies from the cumulative-logit model planned for the real data and analyses each as the real data will be analysed. The procedure stays within the Bayesian framework throughout, because the confirmatory test is a Bayes factor. Its result is the probability that a study reaches a Bayes factor of 10, a different quantity from the power to return a significant p-value.

The per-language simulations take their effects from the pilot. For each language, the planned model is fitted to the pilot ratings. On every simulated study, the fixed effects and ordinal thresholds are drawn together from one draw of that fit's posterior, and the random-effect covariance is held at its posterior mean. New participants and new verb deviations are then drawn around the pilot's own verbs and their measured affectedness scores. Each reported rate is an assurance, a detection rate averaged over the effect sizes the pilot leaves plausible, which allows for the uncertainty of a single pilot's point estimate (Albers & Lakens, 2018). Every pilot language goes forward whatever its pilot showed, so the follow-up bias that Albers and Lakens describe does not arise.

Published effect sizes are subject to publication bias and tend to overstate true effects (Button et al., 2013; Gelman & Carlin, 2014), so design analyses anchored to them can be optimistic (Vasishth et al., 2018). In this report, published estimates enter only the multi-language synthesis, for the four languages the pilot does not cover, and there they carry extra uncertainty to allow for the change of protocol. Their average, -0.34 per standard deviation of affectedness, is close to the three pilots' -0.29, so they show no sign of inflation relative to the CLAPS protocol.

Affectedness is a property of the verb. In the study, it will be rated in a separate norming task, in which twenty further speakers of each language rate the affectedness of the patient in each event, once for each agent-gender version, and a verb's score is the mean of its two version scores. Each verb therefore carries one affectedness score, the same across participants and sentence types, and every affectedness effect is estimated from differences between verbs. The verbs and the spread of

their scores bound the precision of those effects, however many participants take part. This is the stimulus-sampling limit long recognised in psycholinguistics (Clark, 1973) and quantified for designs with crossed random effects by Westfall et al. (2014; see also Brysbaert & Stevens, 2018; Judd et al., 2017). With unlimited participants, the standard error of such an effect cannot fall below the by-verb standard deviation of that effect divided by the square root of the verbs' summed squared affectedness deviations, a limit that can be read from the fitted pilot models without simulation.

The analysis model has the maximal random-effects structure the design supports (Barr et al., 2013), with correlated by-participant slopes for sentence type, affectedness and their interaction, and by-verb intercepts and sentence-type slopes. Because affectedness does not vary within a verb, a by-verb affectedness slope cannot be estimated and is omitted. Affectedness is centred and divided by twice its standard deviation within each language, and effects are reported per standard deviation of affectedness. The priors are centred at zero on the focal slopes, so they favour neither sign. The affectedness slope and the active interaction take normal priors with a standard deviation of 0.5 on the model's scale, or 0.25 per standard deviation of affectedness, and the pseudo-passive interaction takes 0.6, or 0.30 per standard deviation. The other fixed effects take normal priors with a standard deviation of 1.5. The ordinal thresholds take Student-t priors with three degrees of freedom and a scale of 2.5, and the random-effect standard deviations Student-t priors with three degrees of freedom and a scale of 1. Their correlations take an LKJ prior with a shape of 2 (Gelman, 2006; Lewandowski et al., 2009).

Evidence for each hypothesis is a directional Bayes factor, the posterior odds that the coefficient carries the predicted sign divided by the same odds under the prior (Morey & Wagenmakers, 2014). It is estimated from posterior draws and from prior draws sampled alongside them. The two signs it compares are each represented by the zero-centred prior restricted to that side, which is a half-normal. Under these symmetric priors, a Bayes factor of 10 corresponds to a posterior probability of .91 that the effect has the predicted sign, and a Bayes factor of 1/10 or less is the same weight of evidence for the opposite sign. A simulated study counts as a detection when its Bayes factor reaches 10.

Each stimulus exists in two versions, one with a woman agent and one with a man agent. Each participant will see one version of each verb, so an English or Turkish session has 216 sentences, and the simulations describe that session. The versions are balanced across participants by two complementary lists (Appendix B).

The per-language results rest on a dedicated set of 120 simulated studies at each of 70, 80 and 100 participants per language, each scored for both hypotheses, at the pilot's verb counts of 72 in English and Turkish and 66 in Norwegian. The repository's wider grid adds 24 earlier studies at 70 and 100 participants, scored for H1 alone. Each study is fitted with four chains of 3,000 iterations at a target acceptance rate of .99 and a maximum treedepth of 12. After 1,000 warm-up iterations per chain, 8,000 post-warm-up draws remain. The confirmatory analysis will use sixteen chains of 5,000 iterations with 2,000 of warm-up. The shorter chains add Monte-Carlo noise to each simulated

Bayes factor without shifting it systematically. Each per-language rate in Table 1 carries an exact 95% interval. Over 120 studies, a rate near 90% has a Monte-Carlo standard error of about three percentage points, so differences of a few points between neighbouring sample sizes are noise.

The simulations ran on the University of Oxford's Advanced Research Computing service, under R 4.4.2 with brms 2.21.0 and the cmdstanr backend on CmdStan 2.39.0, and the lockfile `renv.lock` fixes the package versions. The analysis code, the simulation settings and the aggregated results are openly available at https://github.com/pablobernabeu/CLAPS-analysis, where H1 and H2 carry the labels h1b and h2b, and this report rebuilds from those results without the raw pilot data, which are not yet public.

Results

Table 1 gives the detection rate for each hypothesis by language and sample size.

Table 1

Detection Rates by Language and Participant Count

Language	Participants	H1	H2	Against H2
English	70	79% [71, 86]	93% [87, 97]	0% [0, 3]
	80	90% [83, 95]	95% [89, 98]	0% [0, 3]
	100	89% [82, 94]	99% [95, 100]	0% [0, 3]
Turkish	70	56% [46, 65]	1% [0, 5]	72% [63, 80]
	80	57% [47, 66]	0% [0, 3]	82% [74, 88]
	100	54% [45, 63]	0% [0, 3]	77% [68, 84]
Norwegian	70	95% [89, 98]	–	–
	80	97% [92, 99]	–	–
	100	98% [93, 99]	–	–

Note. Each row rests on 120 simulated studies. Entries are the percentage of studies reaching a directional Bayes factor of 10, with an exact 95% interval in brackets. H1 is the active-by-affectedness interaction and H2 the pseudo-passive-by-affectedness interaction, both predicted negative. The last column gives the percentage of studies in which the Bayes factor for H2 is 1/10 or less, evidence against it. The Norwegian materials include no pseudo-passive.

H1, the Primary Hypothesis

At 80 participants, H1 reaches a Bayes factor of 10 in 90% [83, 95] of simulated English studies, 97% [92, 99] of Norwegian ones and 57% [47, 66] of Turkish ones. The Turkish pilot interaction is the weakest of the three, at −0.15 logits per standard deviation of affectedness against −0.44 in English and −0.27 in Norwegian.

The rates change little between 80 and 100 participants, as the verb limit predicts, so recruiting beyond 80 would add little in any language. At that limit, averaged over the pilot posterior, the interaction would reach a Bayes factor of 10 in about 97% of English studies, over 99% of Norwegian ones and about 65% of Turkish ones.

H2, the Secondary Hypothesis

In English, the pilot's pseudo-passive-by-affectedness interaction is −0.38 logits per standard deviation of affectedness, with a 95% credible interval from −0.56 to −0.20 and all 8,000 posterior draws below zero. At 80 participants, H2 reaches a Bayes factor of 10 in 95% [89, 98] of simulated studies, more often than H1.

In Turkish, the pilot interaction runs the other way, at +0.16 per standard deviation, with a 95% credible interval from +0.03 to +0.28, so the affectedness slope is steeper for Turkish pseudo-passives than for passives. The simulations draw their effects from that posterior and inherit the reversal. At 80 participants, 82% of simulated Turkish studies give a Bayes factor for H2 of 1/10 or less, evidence against it, and the remaining 22 of 120 are inconclusive.

The negative pseudo-passive interaction also appears in Balinese, Indonesian and Mandarin in the per-language models of the meta-analysis. Those studies inform the planning, but the confirmatory data will be collected under the CLAPS protocol alone.

These per-language rates come from the three pilot languages, and 13 of the 20 planned languages have no data yet. For a language whose effect is drawn from the population the next section assumes, a normal approximation to the per-language test, calibrated against the three pilots' simulated rates, puts the expected detection rate at 63% for H1 and 72% for H2 on the spread across the measured languages (61% to 72% and 49% to 86% across the three spreads considered there).

The Pooled Synthesis

The pooled model tests H1 across all the languages together, and H2 across those with a pseudo-passive. It has the per-language random-effects structure with random slopes by language added, so its population-level interactions estimate the cross-linguistic effects and each language can depart

from them. Each language keeps its own by-verb effects, since a verb and its translations mean similar but never identical things, and verb effects correlate only about 0.2 across the meta-analysis languages. Separate effects also give slightly more power than one shared effect per verb. The pooled model is planned as a synthesis with no confirmatory claim, reported at whatever number of languages takes part, and the confirmatory claims rest on the per-language tests.

With the three pilot languages alone, the pooled model reaches a Bayes factor of 10 on the active interaction in 57% [47, 67] of 107 simulated studies at 50 to 100 participants per language. The rate shows no trend over that range. It is held down by the variation of the effect between languages, which neither verbs nor participants reduce and which three languages estimate poorly. These simulations used the structure of the published meta-analysis model, which also carries a by-verb affectedness slope.

The collaborators expect at least 20 languages, 15 of them with a pseudo-passive, so the synthesis is also simulated with 20 languages. A two-stage version simulates each language's estimate of an interaction and combines the estimates in a Bayesian random-effects meta-analysis. Its one free parameter, the prior that the full model in effect places on the between-language spread, is calibrated against the full model's three-language simulations. The seven languages measured so far take part as CLAPS samples, with effects drawn from their estimates. They are the three CLAPS pilots, and Balinese, Hebrew, Indonesian and Mandarin, the four languages of the meta-analysis other than English. The other 13 are drawn from a population of languages. H2 is simulated in the same way in the 15 languages with a pseudo-passive, five of them among the seven, and Table 2 gives both results.

Table 2

Pooled Detection Rates with Twenty Languages, the Seven Measured Ones Among Them

	H1, active		H2, pseudo-passive	
New languages as different as	Median τ	Detection rate	Median τ	Detection rate
The CLAPS pilots	0.16	92%	0.42	58%
All measured languages	0.39	89%	0.45	90%
The other meta-analysis languages	0.53	76%	0.39	95%

Note. Each rate is the percentage of 4,000 simulated studies in which the pooled directional Bayes factor reaches 10, with 20 languages, 15 of them with a pseudo-passive, and 80 participants and 72 verbs per language. τ is the between-language standard deviation of the interaction per standard deviation of affectedness, given as its posterior median. For H2, each set keeps only its languages with a pseudo-passive: two pilots, five measured languages and three meta-analysis languages. Each meta-analysis language carries an extra standard deviation of 0.15 for the change of protocol, or 0.40 in the first H1 row, whose small spread implies that the earlier estimates carry over poorly.

The spread of the effect between languages decides the result, and the languages measured so far estimate it only roughly. For H1, the three CLAPS pilots differ little, the four other meta-analysis languages differ a great deal, and all seven together lie between. The larger spread may reflect real differences between the languages, or the earlier studies' procedures, which differed in scale length, verb count and affectedness norming. The spread across all seven languages is the planning value used here, since three languages are too few to estimate a spread well and the larger spread may be partly procedural. On that value, the pooled test of H1 reaches a Bayes factor of 10 in 89% of simulated studies, against 92% if new languages differ as little as the pilots and 76% if they differ as much as the meta-analysis languages.

For H2, the spread across the five measured languages with a pseudo-passive is again the planning value, and on it the pooled test reaches a Bayes factor of 10 in 90% of simulated studies. Here the pilots are the unfavourable set. The Turkish reversal offsets the English effect, so new languages drawn around the two average close to zero, and on their spread the rate is 58%.

The Sampling Plan

Every team will collect usable data from 80 participants, replacing any participant who is excluded, a number fixed in advance because each team must state its sample size to its ethics committee. Each language will use as close to 72 verbs as its supply of suitable verbs allows.

Nature Human Behaviour sets its 0.95 power requirement for frequentist analysis plans. For a Bayesian plan, it asks that data collection continue until the Bayes factor is "at least 10 times in favour of the experimental hypothesis over the null hypothesis (or vice versa)". Authors with limited resources may instead state a maximum feasible sample, "sufficiently large that inconclusive results at this sample size would nevertheless be an important message for the field". CLAPS takes the second route for both hypotheses. Each team stops at 80 participants, the most the collecting teams can provide, and a test whose Bayes factor has reached neither 10 nor 1/10 by then is reported as inconclusive. At that sample, each language is close to the limit its verbs set, so further participants would add little to the chance of reaching the threshold, and an inconclusive result would still reflect an estimate with nearly the precision the language's verbs allow. Table 3 summarises the planned analyses and their detection rates.

Table 3

Planned Analyses and Their Detection Rates

Analysis	Coefficient and direction	Test	Detection rate
H1, primary	Active by affectedness, negative	Directional Bayes factor of 10, in each language	English 90%, Norwegian 97% and Turkish 57%
H2, secondary	Pseudo-passive by affectedness, negative	Directional Bayes factor of 10, in each language with a pseudo-passive	English 95%. In Turkish, 82% give evidence against it
Pooled, H1	Active by affectedness across languages, negative	Directional Bayes factor, reported as a synthesis with no confirmatory claim	57% with the three pilot languages, and 89% with 20 including the seven measured ones
Pooled, H2	Pseudo-passive by affectedness across languages, negative	As above, in the languages with a pseudo-passive	90% with the 15 of those 20 that have one

Note. The per-language rates are at 80 participants. The three-language pooled rate combines simulations at 50 to 100 participants per language. The 20-language rates are at 80 participants per language, on the spread across the measured languages that carry the interaction.

The Bayes factor to be registered is directional (Morey & Wagenmakers, 2014). Each hypothesis is represented by the half-normal given in the Design-Analysis Method, one of the forms the journal names for representing a prediction, and its competitor is the reversed effect, represented by the same half-normal on the other side. A Bayes factor of 10 is therefore strong evidence for the hypothesis over its reversal, and one of 1/10 or less is equally strong evidence against it.

Turkish follows the same route. Its H1 interaction reaches a Bayes factor of 10 in 57% of simulated studies and in at most about 65% with unlimited participants, so an inconclusive Turkish result would be informative for the reason given above, and its pilot points against H2. Turkish is tested on both hypotheses like every other language, since leaving a language out on the strength of its pilot would amount to choosing languages by their results.

Limitations

At 80 participants, no simulated fit has a divergent transition or a saturated treedepth. Confirmatory fits must also reach an R-hat below 1.01 and bulk and tail effective sample sizes of at least 400.

With the shorter runs used for simulation, 27 of 120 English fits meet those thresholds, as do 33 Norwegian and 12 Turkish fits. Every rate counts all fits. Over the passing fits alone, the rates are 81% in English, 100% in Norwegian and 50% in Turkish, each within what a random subset of that size would show. The confirmatory chains will also be half as long again, and sixteen of them will give six times the post-warm-up draws.

Fitted to the combined pilot data, 75,528 ratings over 210 verbs, the pooled model estimated the interaction at about -0.24 logits per standard deviation of affectedness, with a 95% credible interval from -0.51 to $+0.15$. That estimate is close to the average of the three pilot estimates, -0.29. The pooled model is harder to fit than the per-language ones, however. It saturated the maximum treedepth on three quarters of its iterations, and confirmatory fits must show none. A pooled fit to the collected data that misses the criteria falls back to the two-stage synthesis of the per-language estimates, the analysis behind the 20-language rates above.

The 20-language figures rest on an assumed population of languages. The seven measured so far are a convenience sample from five families in two macroareas, and the simulations treat the other languages as draws from a population like them.

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Appendix A

Composition of the Pilot Sample

This appendix describes the pilot sample whose ratings give the descriptive trends in Figure 1 and to which the planned model is fitted for the simulations. Table A1 gives the counts, read directly from the harmonised pilot data.

Table A1

Composition of the Harmonised Pilot Sample, by Language

Language	Participants	Verbs	Sentence types	Items	Judgements
English	70	72	3	144	30,240
Turkish	70	72	3	144	30,240
Norwegian	57	66	2	132	15,048

Note. Items are verb-by-agent-gender combinations, so the item count is twice the verb count. Every pilot participant saw both versions of every verb, whereas in the study itself each participant will see one version of each verb (Appendix B). The Norwegian materials include no pseudo-passive, hence only two sentence types. Norwegian sessions also included a synthetic passive, which the preregistered analysis excludes, so a Norwegian participant rated one and a half times as many sentences as this table counts.

Appendix B

The Agent-Gender Lists

This appendix sets out how the two agent-gender versions of each stimulus are balanced across participants, and why. Each participant will see one version of each verb, so the balance has to come from the assignment of versions to participants.

Balance matters because agent gender moves the interaction in English. In a linear mixed model of the 1-to-7 pilot ratings, used here only to gauge the relative size of the gender term, with agent gender coded plus or minus a half, the English interaction is -0.16 rating points per standard deviation of affectedness and its gender-by-active-by-affectedness term is $+0.068$ ($SE = 0.023$). The interaction is therefore markedly stronger for man-agent items than for woman-agent ones. A design in which every participant saw only woman-agent versions, or only man-agent ones, would move the English interaction by about 21%, against 1% in Turkish and 3% in Norwegian.

Two complementary lists remove that shift without lengthening the session. The verbs are split into two sets of 36, or 33 in Norwegian, matched on affectedness and verb class, and the participants into two lists. List 1 shows the first set with woman agents and the second with man agents, and list 2 the reverse. Each participant therefore sees both agent genders equally often, each verb is rated in each version by half the sample, and the estimand remains the slope averaged over agent gender. Because both lists show both agent genders on matched verb sets, an unequal split between the lists leaves the interaction essentially unchanged. Even so, assigning participants to the lists alternately, and replacing exclusions within the same list, keeps each verb's two versions evenly rated. Splitting the agent-gender versions across participants in this way follows Ambridge et al. (2016, Study 3) and the Hebrew study of Ambridge et al. (2023).

Both versions stay in the affectedness rating task, which asks for one judgement per verb and agent gender and is short. Halving it would leave each agent-specific score resting on half as many raters and would attenuate every slope estimated from those scores.